

Aladdin

Aladdin - nettitude, GitHub.

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- Repository: <https://github.com/nettitude/Aladdin>
- Commit: e47477cfd2d85bb738756364d96fb1bac9eaf399
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LICENSE

Blob 1e54c17f6250 , 1557 bytes, at commit e47477cfd2d8 .

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README.md

Blob `f6dfb60ea77f`, 4308 bytes, at commit `e47477cfd2d8`.

Aladdin

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Lefty @l3f7er1span - Nettitude Red Team - 2022 / 2023

About

Aladdin is a payload generation technique based on the work of James Forshaw (@tiraniddo) that allows the deserialization of a .NET payload and execution in memory.

The original vector was documented on <https://www.tiraniddo.dev/2017/07/dg-on-windows-10-s-executing-arbitrary.html>.

By spawning the process `AddInProcess.exe` with arguments `` `/guid:32a91b0f-30cd-4c75-be79-ccbd6345de99`` and `` /pid:``, the process will start a named pipe under `\\.\pipe\32a91b0f-30cd-4c75-be79-ccbd6345de99`` and will wait for a .NET Remoting object. If we generate a payload that has the appropriate packet bytes required to communicate with a .NET remoting listener we will be able to trigger the `ActivitySurrogateSelector` class from `System.Workflow.ComponentModel`. and gain code execution.

Originally, James Forshaw released a POC at

`` https://github.com/tyranid/DeviceGuardBypasses/tree/master/CreateAddInIpcData ``.

However this POC will fail on recent versions of Windows since Microsoft went ahead and patched the vulnerable `System.Workflow.ComponentModel` (<https://github.com/microsoft/dotnet-framework-early-access/blob/master/release-notes/NET48/dotnet-48-changes.md>).

Nick Landers (@monoxgas) however, identified a way to disable the check that Microsoft introduced and wrote a detailed article at

<https://www.netspi.com/blog/technical/adversary-simulation/re-animating-activitysurrogateselector/> . The bypass is documented at <https://github.com/pwntester/ysoserial.net/pull/41> .

Aladdin is a payload generation tool, which using the specific bypass as well as the necessary header bytes of the .NET remoting protocol is able to generate initial access payloads that abuse the `AddInProcess` as originally documented.

The provided templates are:

- HTA
- VBA
- JS
- CHM

Notes

In order for the attack to be successful the .NET assembly must contain a single public class with an empty constructor to act as the entry point during deserialization. An example assembly has been included in the project.

```
public class EntryPoint {  
    public EntryPoint() {  
        MessageBox.Show("Hello");  
    }  
}
```

Usage

Usage:

<code>-w, --scriptType=VALUE</code>	Set to js / hta / vba / chm.
<code>-o, --output=VALUE</code>	The generated output, e.g: <code>-o C:\Users\Nettitude\Desktop\payload</code>
<code>-a, --assembly=VALUE</code>	Provided Assembly DLL, e.g: <code>-a C:\Users\Nettitude\Desktop\popcalc.dll</code>
<code>-h, --help</code>	Help

OpSec

- The user supplied .NET binary will be executed under the `AddInProcess.exe` that gets spawned from the HTA / JS payload. The spawning of the processes currently happens using the 9BA05972-F6A8-11CF-A442-00A0C90A8F39 COM object (<https://dl.packetstormsecurity.net/papers/general/abusing-objects.pdf>) which will launch the process as a child of `Explorer.exe` process.
- The GUID supplied in the process parameters of `AddInProcess.exe` can be user controlled. At the moment the guid is hardcoded in the template and the code.
- CHM executes the JScript through XSLT transformation

Defensive Considerations

- `Addinprocess.exe` will always launch with `/guid` and `/pid`. Baseline your environment for legitimate uses - monitor the rest

Useful References:

- <https://www.tiraniddo.dev/2017/07/dg-on-windows-10-s-executing-arbitrary.html>
- <https://www.netSPI.com/blog/technical/adversary-simulation/re-animating-activitysurrogateselector/>

Readme / Credits

Code is based on the following repos:

- <https://github.com/tyranid/DeviceGuardBypasses/tree/master/CreateAddInIpcData>
- <https://github.com/pwntester/ysoserial.net>

Shouts to:

- @m0rv4i for helping with C# nuances
- @ace0fspad3s for troubleshooting
- @ Nettitude RT for being awesome

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